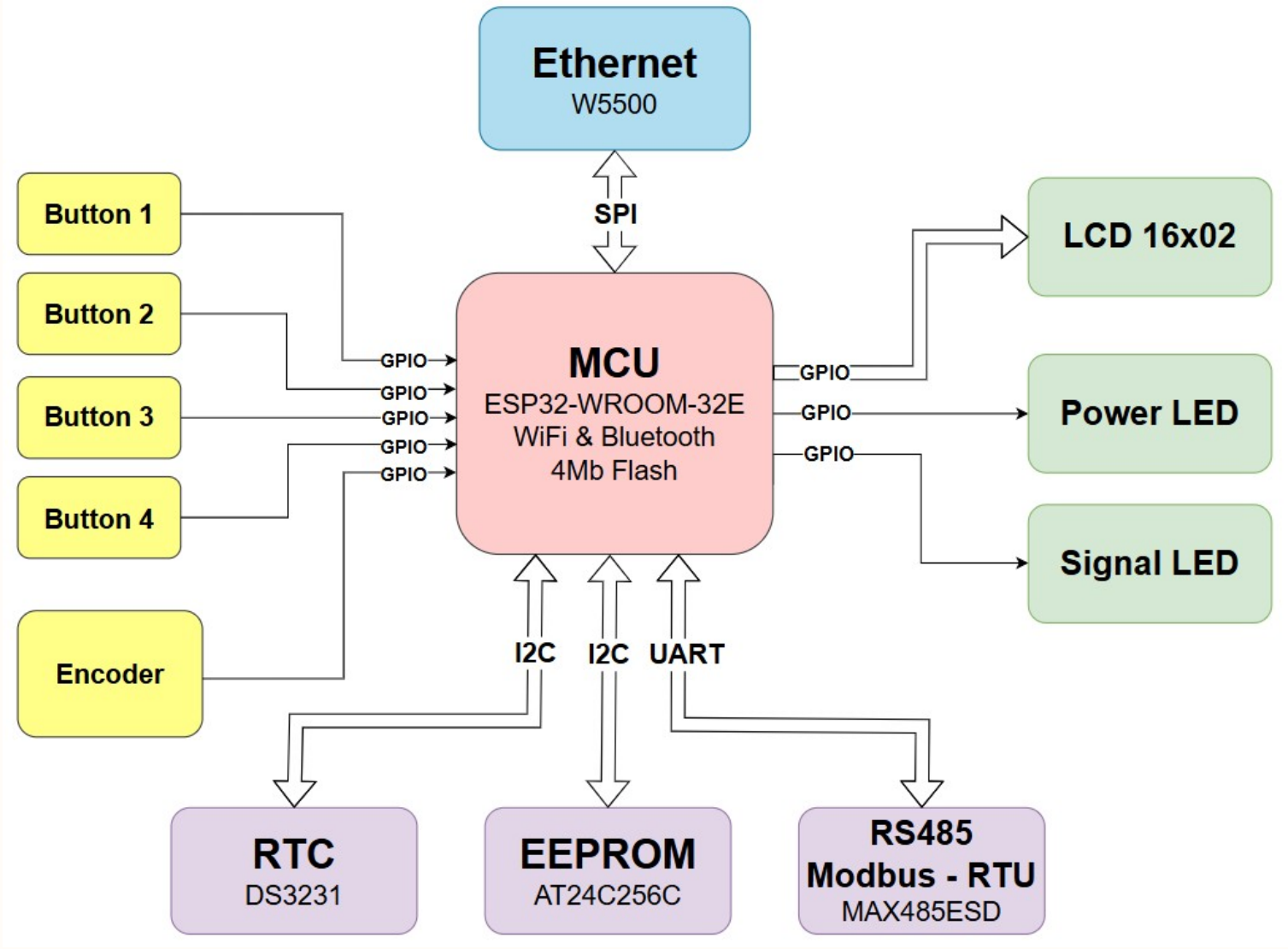
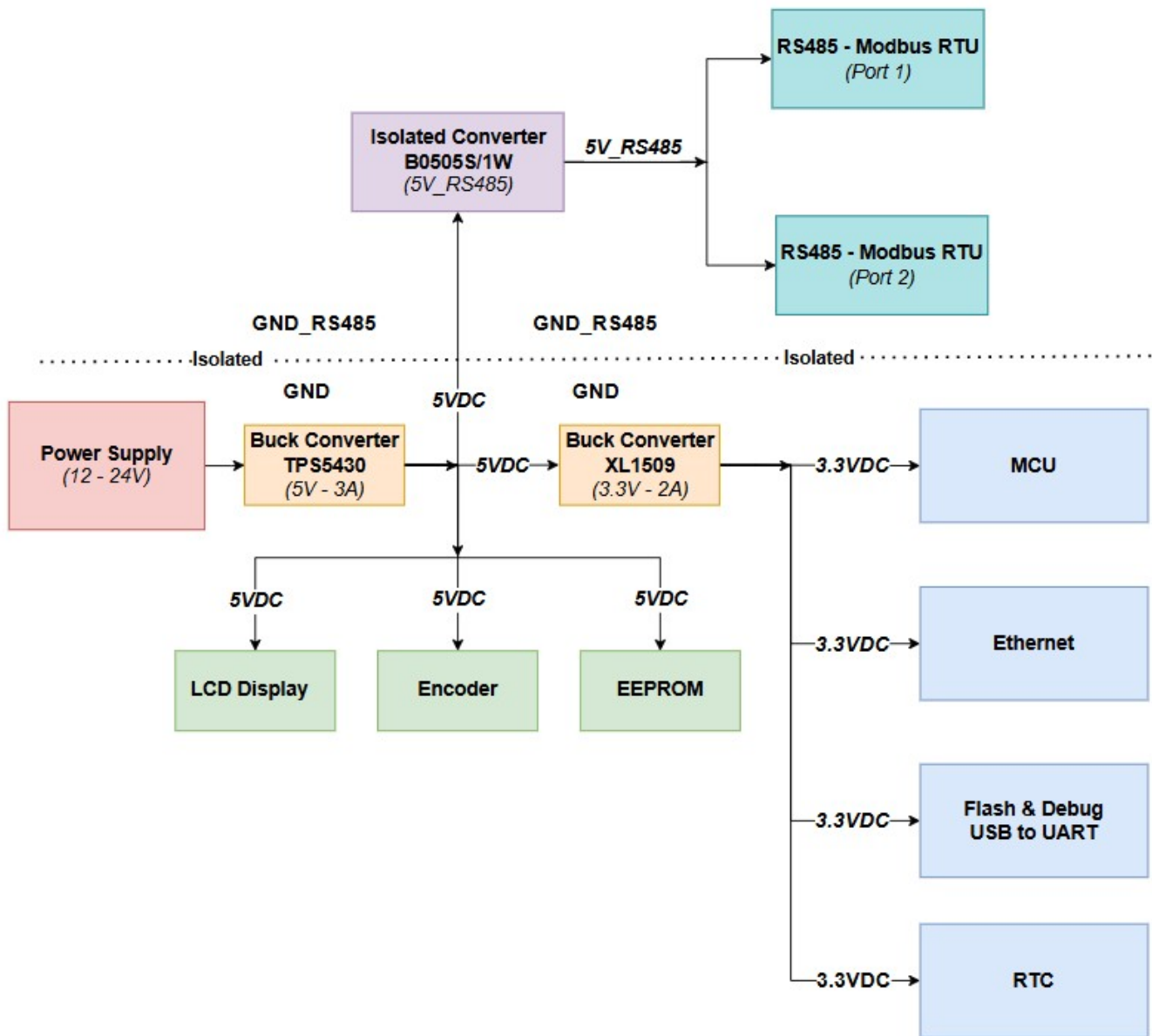


1		2		3		4	
A							
B							
C							
D							
1		2		3		4	

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File:	D:\08 Semester 8\..Index Page.SchDoc		Drawn By:



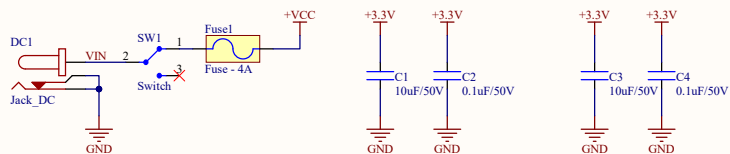
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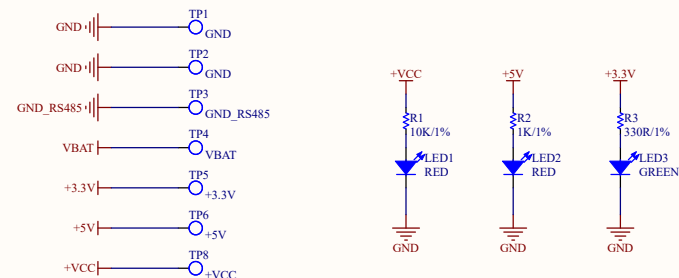
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Date: 2/25/2026	Sheet of	
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DC JACK & FILTER

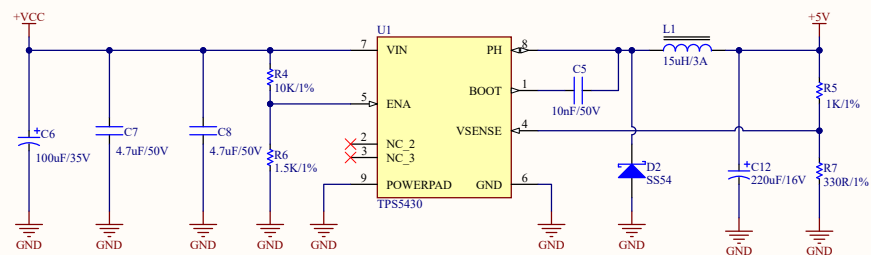
DC IN 12 - 24V



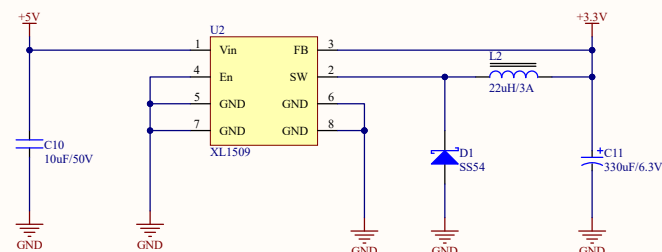
TEST POINT & POWER LEDs



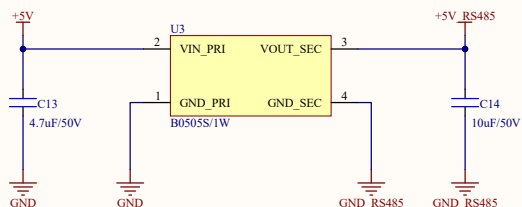
BUCK - CONVERTER - 5V



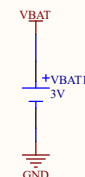
BUCK - CONVERTER - 3.3V



ISOLATED - CONVERTER 5V_RS485

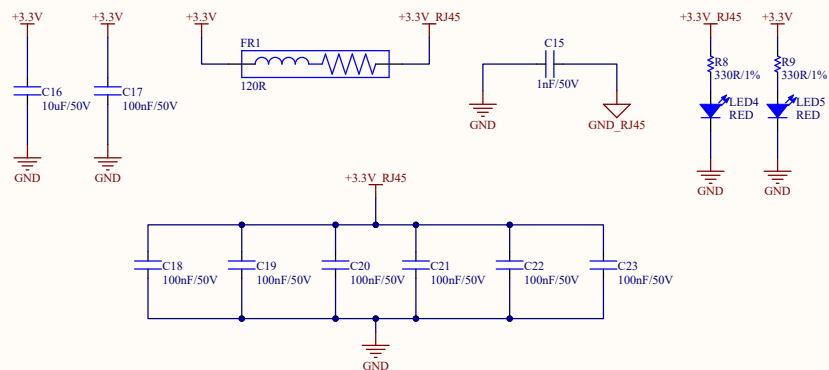


PIN 3V FOR RTC

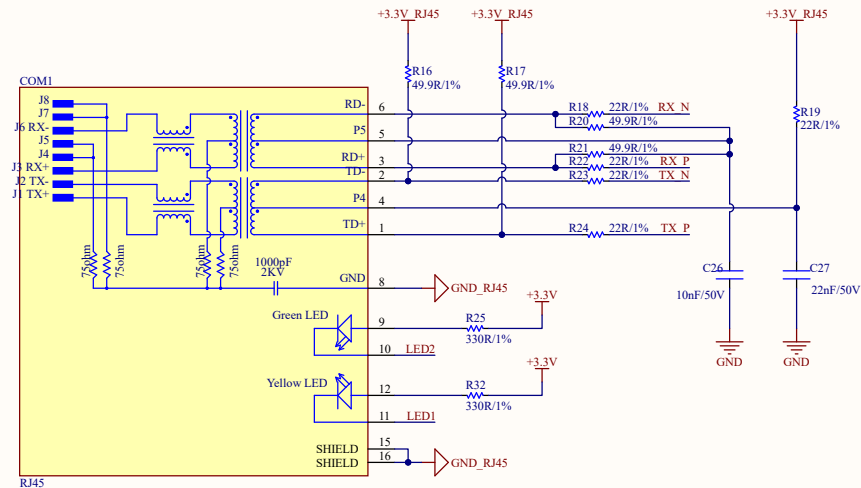


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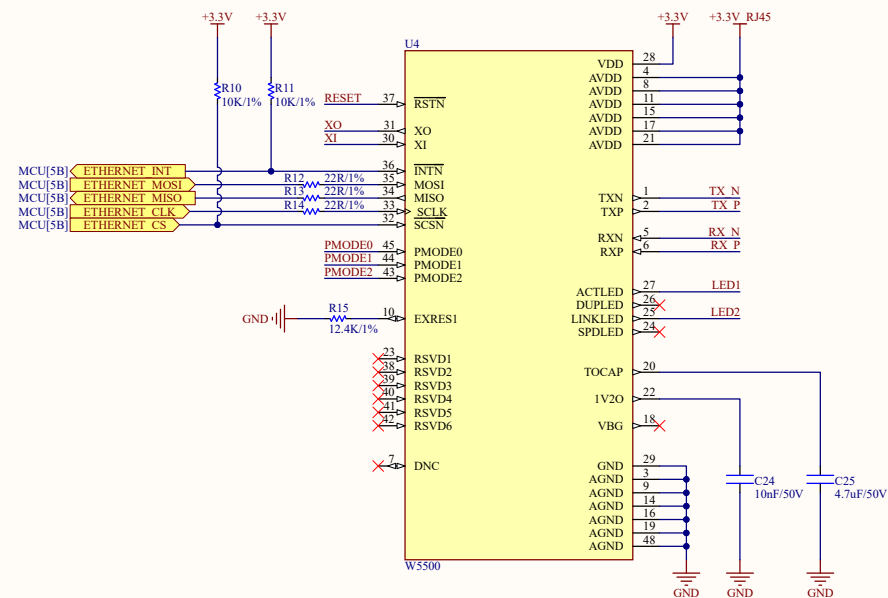
POWER FILTERs & POWER LEDs



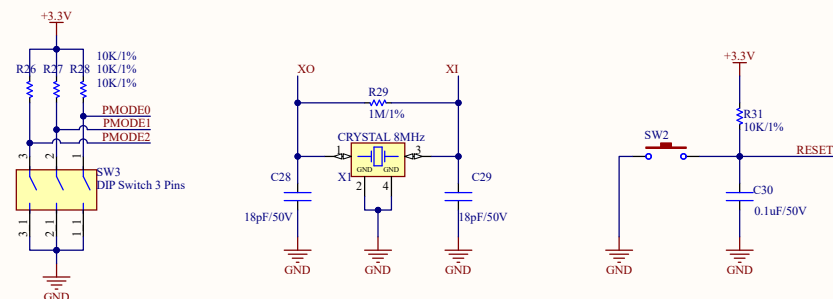
RJ45 PORT



ETHERNET - W5500



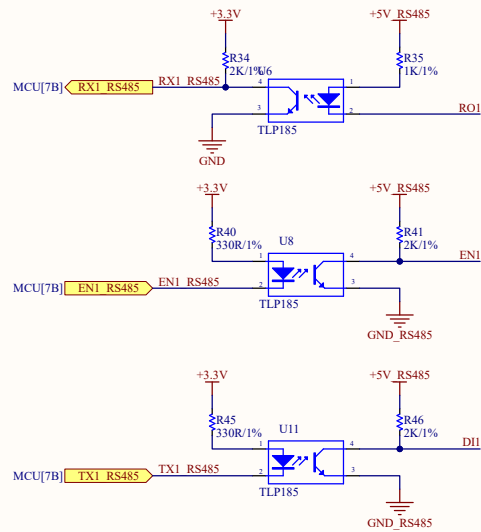
MODE SWITCH & CRYSTAL & RESET BUTTON



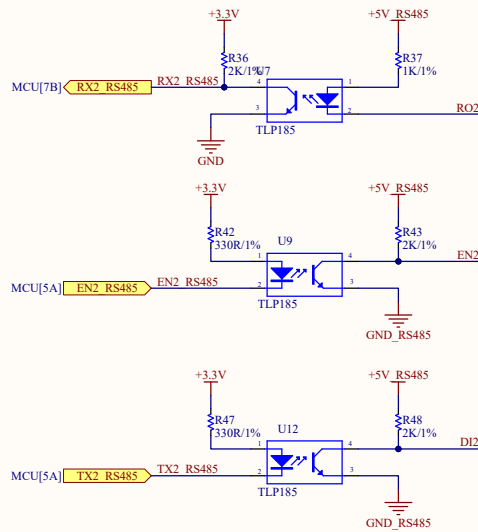
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COMMUNICATE WITH MCU

PORT 1



PORT 2

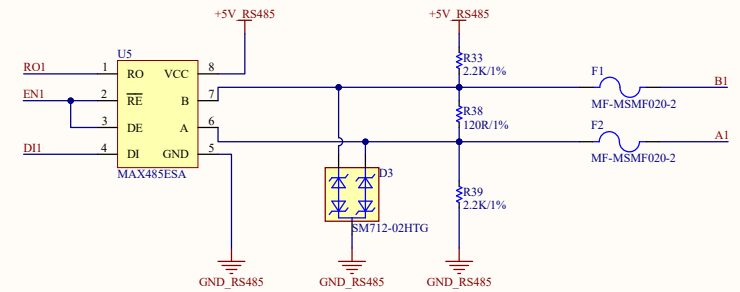


CONNECTORs & TEST POINTs

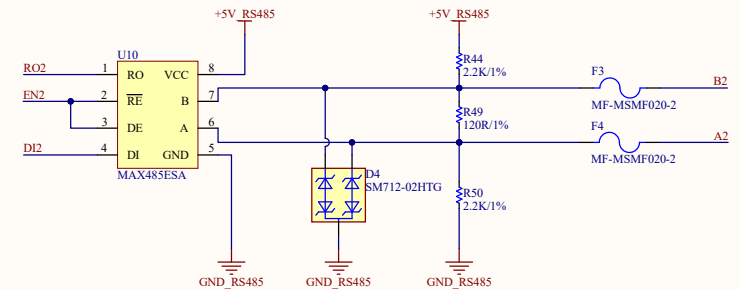


RS485 PORTS

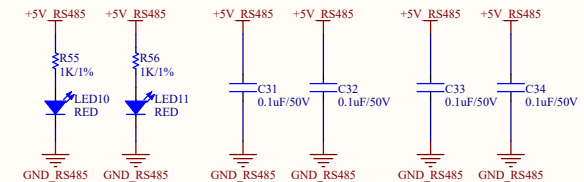
PORT 1



PORT 2

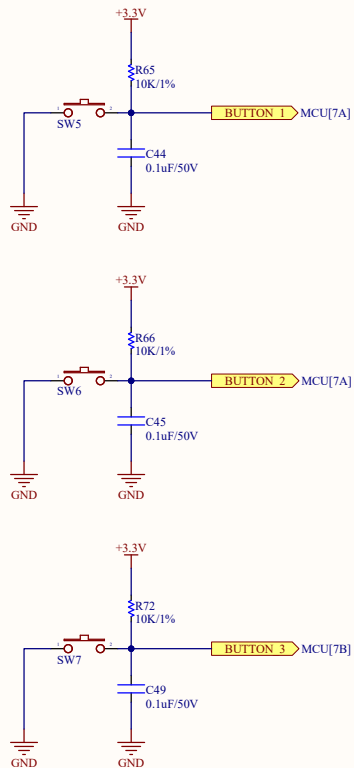


POWER LEDs & POWER FILTERs

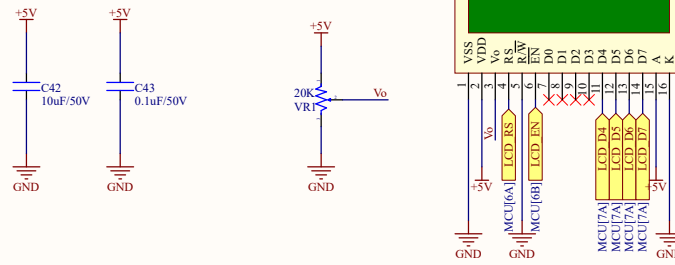


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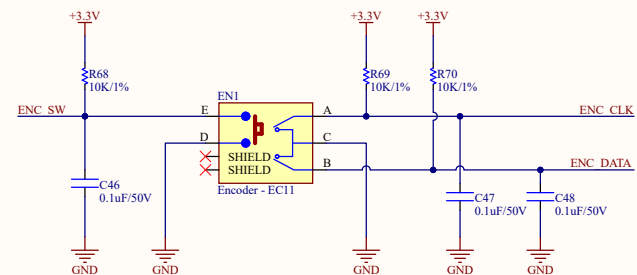
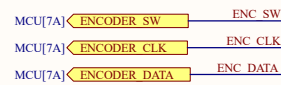
USER BUTTON



LCD 16x4A

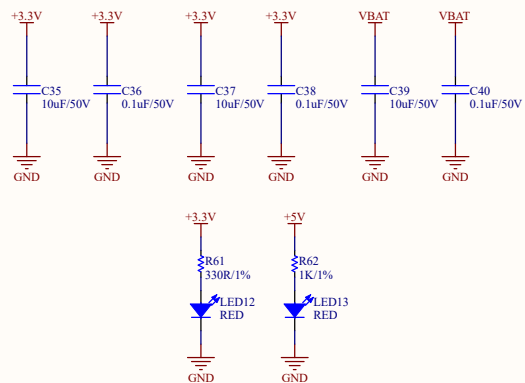


ENCODER

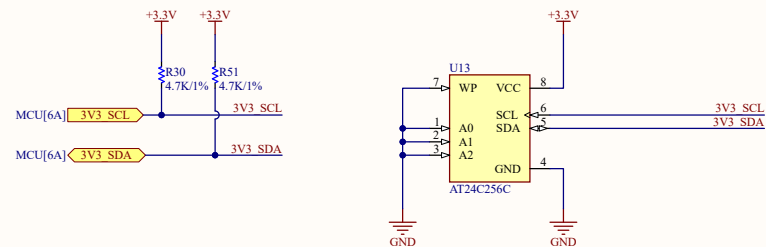


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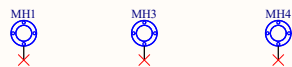
POWER FILTERs & POWER LEDs



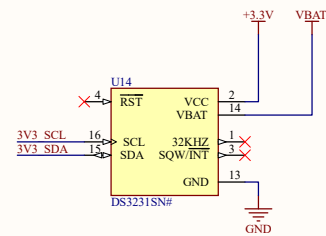
EEPROM MEMORY



MOUNTING HOLES



REAL TIME CLOCK (RTC)



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File:	D:\08 Semester 8\...RTC and EEPROM	Author By:

[illegible]

Pin connection diagram for ESP32-S3-WROOM-1U-NR2 module. The diagram shows the module's pins (1-40) connected to various components. Key connections include:

- Power:** 3V3 to EN, IO20, and IO48; GND to GND pins.
- RS485:** RS485 Port[3B] to TX2 and RX2; RS485 Port[3B] to TX2 and RX2.
- RTC and EEPROM:** RTC and EEPROM[4B] to SDA, SCL, and CS; Ethernet[5B] to ETH_INT, ETH_MOSI, ETH_CLK, ETH_MISO, and ETH_CS.
- LCD:** LCD D4-D7 to LCD D4-D7.
- Buttons:** BUTTON 1 and 2 to BUTTON 1 and 2; BUTTON 3 to BUTTON 3.
- Encoders:** ENCODER_DATA to ENCODER_DATA.
- RS485 Ports:** RX1 and TX1 RS485 to RX1 and TX1; RX2 and TX2 RS485 to RX2 and TX2.

The module is labeled ESP32-S3-WROOM-1U-NR2.

Strapping Pin	Default Configuration	Bit Value
GPIO0	Weak pull-up	1
GPIO3	Floating	–
GPIO45	Weak pull-down	0
GPIO46	Weak pull-down	0

[illegible]

Auto program				
DTR	RTS-->	EN	IO0	IO2
1	1	1	1	1
0	0	1	1	1
1	0	0	1	1
0	1	1	0	0

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File:	D:\08 Semester 8\MCU.SchDoc	Drawn By: